



Predicting House Prices using Linear Regression

Tools Used:

- Python (Google Colab)
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn
- GitHub

PHASE 1: Understanding the Dataset

- 545 rows and 13 columns
- Target variable:
- **Price** (house sale price in INR)

Numerical Columns	Categorical Columns
1. Price 2. Area 3. Bedrooms 4. Bathrooms 5. Stories 6. Parking	7. Main Road 8. Guest Room 9. Basement 10. Hot Water Heating 11. Air Conditioning 12. Preferred Area 13. Furnishing Status

PHASE 2: Data Loading and Cleaning

Loaded the dataset (Housing.csv) into Google Colab and performed the following cleaning steps:

- No missing values found — `df.isnull().sum()` returned all zeros
- No duplicate rows detected — `df.duplicated().sum()` returned 0
- Encoded binary categorical columns (yes/no) into 1/0
- Applied One-Hot Encoding on furnishing status (semi-furnished, unfurnished)

PHASE 3: Descriptive Statistics

Summary Statistics (Price column):

Mean Price	₹ 4,766,729
Std Deviation	₹ 1,870,440
Min Price	₹ 1,750,000
Max Price	₹ 13,300,000
Median (50th %ile)	₹ 4,340,000

- High standard deviation indicates a wide spread in house prices
- Mean is significantly higher than the median, suggesting right skew in price distribution

PHASE 4: Exploratory Data Analysis (EDA)

1. Correlation Heatmap

Generated a heatmap using `sns.heatmap()` on numeric columns. Area, bathrooms, and stories showed the strongest positive correlations with price.

2. Price Distribution (Histogram + KDE)

The histogram showed that most house prices are concentrated in the lower range, with a right-skewed distribution — a few premium properties drive the mean upward.

3. Area vs Price (Scatter Plot)

A positive linear trend was visible between area and price, confirming area as a strong predictor variable.

PHASE 5: Data Preprocessing

Feature Engineering:

- Dropped the target variable (price) to create the feature matrix X
- Kept price as the target vector y

Encoding:

- Binary columns (mainroad, guestroom, basement, hotwaterheating, airconditioning, prefarea) mapped from yes/no to 1/0
- furnishingstatus encoded via `pd.get_dummies()` with `drop_first=True` to avoid multicollinearity

Feature Scaling:

- Applied `StandardScaler` on numerical columns: area, bedrooms, bathrooms, stories, parking
- Scaling ensures all features contribute proportionally to the model

Handling Missing Values (Post-Encoding):

- Used SimpleImputer (strategy='mean') to handle any NaN values in the feature matrix before model training

PHASE 6: Model Building — Linear Regression

Train-Test Split:

- 80% training data → 436 rows
- 20% testing data → 109 rows
- random_state=42 used for reproducibility

Model:

- Algorithm: LinearRegression() from sklearn.linear_model
- Fit on imputed training data (X_train_imputed, y_train)
- Predicted house prices on X_test_imputed

PHASE 7: Model Evaluation

Metric	Value
R-squared (R^2)	0.6529
Mean Absolute Error (MAE)	970,043.40
Mean Squared Error (MSE)	1,754,318,687,330.67

Interpretation: The R^2 score of 0.65 means the model explains ~65% of the variance in house prices. The MAE of ~970K indicates an average prediction error of roughly ₹9.7 lakh.

PHASE 8: Data Visualization

4. Correlation Heatmap

Highlighted that area, bathrooms, and stories are the most correlated features with price. Parking had a weaker correlation.

5. Price Distribution Histogram (with KDE)

Showed right-skewed distribution of house prices. Most houses fall in the ₹1.75M–₹5M range.

6. Area vs Price Scatter Plot

Positive linear relationship confirmed. Larger area generally commands a higher price.

7. Actual vs Predicted Price Scatter Plot

Points clustered around the red diagonal (perfect prediction line), indicating the model performs reasonably but has room for improvement at extreme values.

PHASE 9: Recommendations & Insights

- Area is the strongest predictor of house price — larger properties consistently fetch higher prices.
- Amenities like air conditioning, preferred area location, and number of bathrooms significantly push prices upward.
- Furnishing status matters — furnished homes tend to be priced higher than semi-furnished or unfurnished ones.
- The model achieves a reasonable R^2 of 0.65, but the high MAE suggests that non-linear models (e.g. Random Forest, Gradient Boosting) could better capture complex interactions.
- Right-skewed price distribution suggests applying log transformation on the target variable could improve model performance.
- Feature importance analysis with tree-based models would help identify which variables contribute most to pricing.